Circles Payment Gateway Audit

Gnosis Ltd - Report by Côme du Crest

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Circles Payment Gateway Audit

This document presents the findings of a smart contract audit conducted by Côme du Crest for Gnosis Ltd.

Scope

The scope includes all contracts within `circles-payment-gateway-contracts` as of commit `0x1b35947`.

Context

The payment gateway is a simple contract that emits a `PaymentReceived` event through its factory when receiving tokens before forwarding them to its owner. The event is emitted by the factory likely for easier indexing.

The gateway has two functions to set and remove trusts to any address on the hub. The functions can only be called by the owner set during deployment. Otherwise, it implements the standard `onERC1155Received()` and `onERC1155BatchReceived()` that verify that it has been called by the circles hub and call `factory.emitPaymentReceived()`.

The factory has a function to create a gateway with `hub` set as an immutable address and `msg.sender` as owner. Only gateways created by the factory can call functions to emit `PaymentReceived` and `TrustUpdated` events. As such, it can be trusted that the `PaymentReceived` events are only emitted when circles transfer are received by a gateway. Similarly, the `TrustUpdated` events can only be emitted when gateway trusts have been updated.

Status

No issues have been found.